

Telecom Customer Churn & Retention Analysis

End-to-End Enterprise Technical & Strategic Intelligence Report

Domain: Telecommunications Analytics	Primary Key: CustomerID	Architecture: Normalized Star Schema (14 Tables)
Dataset Size: 7,043 Customer Profiles	Overall Churn Rate: 26.54% (1,869 Churned)	Toolstack: Power Query, Power Pivot, DAX, Dashboard

1. Executive Summary

Customer attrition represents a major operational and financial challenge in the telecommunications industry. This project delivers an end-to-end data analytics system designed to evaluate customer retention dynamics, isolate root causes of attrition, and provide data-backed strategic recommendations. By converting a flat raw transactional file into a **14-table Relational Star Schema Data Model**, executing exploratory multi-dimensional analysis, and constructing an interactive dashboard, this study delivers actionable insights to stabilize revenue streams and increase customer lifetime value.

7,043	1,869	26.54%	14
TOTAL BASE CUSTOMERS	TOTAL CHURNED CLIENTS	BASELINE CHURN RATE	RELATIONAL MODEL TABLES

2. Data Ingestion, Cleaning & Relational Data Modeling

To preserve data integrity and conform to enterprise database standards, the raw Telco_Customer_Churn dataset was ingested into **Power Query** and normalized into 14 separate entity tables. This architecture prevents data duplication, ensures optimal indexing, and establishes clean foreign-key relationships to CustomerID.

#	Entity Name	Classification	Core Attributes / Key Fields Included
1	Dim_Customers	Dimension	CustomerID, Gender, Senior Citizen, Partner, Dependents
2	Dim_Locations	Dimension	CustomerID, Country, State, City, Zip Code, Lat Long
3	Dim_Contracts	Dimension	CustomerID, Contract Type (Month-to-month, One year, Two year)
4	Dim_PaymentMethods	Dimension	CustomerID, Payment Method (Electronic check, Bank transfer, etc.)
5	Dim_PhoneServices	Dimension	CustomerID, Phone Service, Multiple Lines
6	Dim_InternetServices	Dimension	CustomerID, Internet Service (DSL, Fiber optic, No)
7	Dim_OnlineSecurity	Dimension	CustomerID, Online Security Add-on Status
8	Dim_OnlineBackup	Dimension	CustomerID, Online Backup Service Status
9	Dim_DeviceProtection	Dimension	CustomerID, Hardware Device Protection Plan Status
10	Dim_TechSupport	Dimension	CustomerID, Dedicated Tech Support Plan Status
11	Dim_StreamingTV	Dimension	CustomerID, TV Streaming Subscription Status
12	Dim_StreamingMovies	Dimension	CustomerID, Movie Streaming Subscription Status
13	Fact_MonthlyBilling	Fact Entity	CustomerID, Tenure Months, Monthly Charges, Total Charges, Tenure_Group
14	Fact_CustomerChurn	Fact Entity	CustomerID, Churn Label, Churn Value, Churn Score, CLTV, Churn Reason

3. Exploratory Data Analysis & Key Findings

Multi-dimensional pivot table analysis revealed clear operational and commercial drivers behind customer attrition across contracts, financial channels, technical products, and geographical distribution.

A. Contract Duration vs. Churn Rate

Contract Structure	Retained Customers (No)	Churned Customers (Yes)	Total Subscribers	Segment Churn Rate (%)
Month-to-month	2,220	1,655	3,875	42.71%
One year	1,307	166	1,473	11.27%
Two year	1,647	48	1,695	2.83%
Grand Total	5,174	1,869	7,043	26.54%

Key Insight: Month-to-month contracts experience a massive **42.71% churn rate**, representing 88.5% of total churned users (1,655 out of 1,869). Long-term commitment dramatically improves retention, reducing churn to 2.83% for two-year subscribers.

B. Payment Channels vs. Churn Dynamics

Payment Method	Retained (No)	Churned (Yes)	Total Subscribers	Segment Churn Rate (%)
Electronic Check	1,294	1,071	2,365	45.29%
Mailed Check	1,304	308	1,612	19.11%
Bank Transfer (Automatic)	1,286	258	1,544	16.71%
Credit Card (Automatic)	1,290	232	1,522	15.24%

Key Insight: Customers utilizing **Electronic Checks** exhibit an alarming **45.29% churn rate**. Friction in manual monthly payment portals combined with lack of automated billing accounts for this disproportionate loss compared to auto-debit channels (~15-16%).

C. Service Attach Rates & Support Protection Analysis

Subscribed Service Category	No Add-on (Churn Rate %)	Active Add-on (Churn Rate %)	Key Takeaway
Internet Type (Fiber Optic)	15.24% (DSL: 18.96%)	41.89% (Fiber Optic)	Fiber optic users churn at double the rate of DSL due to uncompetitive pricing or service outages.
Tech Support	41.64% (No Tech Support)	15.17% (Active Tech Support)	Providing dedicated tech support reduces customer attrition by nearly 64%.
Online Security	41.77% (No Security)	14.61% (Active Security)	Security add-ons significantly increase product stickiness and account longevity.

D. Top Root Causes of Churn & Geographic Exposure

- **Primary Attrition Reasons:** Competitive pressure is the primary driver. Key reasons include *Competitor made better offer* (140 clients), *Competitor offered higher download speeds* (189 clients), *Attitude of support person* (192 clients), and *Attitude of service provider* (135 clients).
- **Top Churn Hotspot Cities:** High absolute loss counts are concentrated in major metropolitan centers, led by **Los Angeles** (90 churned out of 305; 29.5%), **San Diego** (50 churned out of 150; 33.3%), **San Francisco** (31 churned; 29.8%), and **Sacramento** (26 churned; 24.1%).

4. Interactive Dashboard Architecture & Visual Design

The interactive analytics dashboard was engineered to convert raw transactional figures into visual executive intelligence. The visual layout follows strict design principles:

- **Executive KPI Cards:** Prominently display top-level metrics (Total Base Customers, Churn Count, Baseline Churn Rate, and Average Charges) at the top of the interface for immediate situational awareness.
- **Comparative Bar / Column Charts:** Used to evaluate category-level churn variance across contract types, payment channels, and internet service subscriptions.
- **Charge Bins & Tenure Histograms:** Discretized continuous monthly charges into structured pricing bins and tenure cohorts to identify price-sensitivity thresholds.
- **Cross-Filtering Slicers:** Dynamic slicers (Contract Type, Senior Citizen, Partner, Churn Label) allow analysts to slice data instantly (e.g., isolating married users on monthly contracts to evaluate family pressure risks).

5. Data-Driven Strategic Recommendations

1. Contract Incentive & Long-Term Migration Programs:

Launch promotional incentives (e.g., first 2 months at 20% discount or free speed upgrades) targeting Month-to-Month subscribers to convert them into 1-Year or 2-Year contracts. Transitioning just 20% of monthly users to annual plans would prevent hundreds of cancellations per quarter.

2. Electronic Check Friction Reduction & Auto-Pay Incentives:

Introduce a \$5/month billing credit for customers who switch from Electronic Checks to automated Direct Debit or Credit Card auto-pay. Eliminating manual billing friction directly targets the 45.29% churn rate in this segment.

3. Support & Security Value-Added Bundling:

Package Tech Support and Online Security into standard internet plans at a discounted bundle rate rather than selling them as optional add-ons. Subscribers with active support and security features exhibit a sub-15% churn rate.

4. Regional Network Optimization & Competitor Defense in High-Churn Cities:

Deploy targeted competitive retention offers in Los Angeles, San Diego, and San Francisco. Concurrently, conduct operational audits on local customer support personnel and Fiber Optic infrastructure reliability in these key cities to address service quality complaints.